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Course: SOC Analyst Tier 1 Graduation Project

Subject: SIEM Introduction Room (LetsDefend) Report

Objective

The objective of this report is to analyze the fundamental architecture and operational lifecycle of a Security Information and Event Management (SIEM) platform. This includes evaluating the market landscape of SIEM products, understanding various log ingestion methods, and demonstrating the critical role of log aggregation, storage, and alerting in modern SOC environments.

SIEM introduction

SIEM, which stands for Security Information and Event Management, is a centralized security platform that gathers data from across an organization's infrastructure, analyzes it, and identifies potential threats in real time. It is an essential tool for SOC analysts as it allows them to monitor security events continuously without manually checking each system individually.

SIEM products

According to the Gartner Magic Quadrant report (June 2021), SIEM solutions in the market are classified into four categories based on two criteria: the vendor's completeness of vision and their ability to execute. The **Leaders** quadrant contains vendors that excel in both areas, making them the most widely adopted enterprise solutions. The **Challengers** quadrant includes vendors with strong execution capabilities but a less comprehensive vision. The **Visionaries** quadrant represents vendors with innovative and forward-thinking ideas but who have yet to fully prove their ability to deliver at scale. Finally, the **Niche Players** quadrant includes vendors that score lower on both criteria, typically serving specific industries or smaller organizations with more limited security requirements.



Source: Gartner (June 2021)

Figure 1: Gartner Magic Quadrant report (June 2021)

Leaders vs Challengers vs Visionaries vs Niche players

Quadrant	Description	Examples
Leaders	High vision & strong execution. Best overall choice for enterprises.	Exabeam, IBM, Securonix, Splunk, Rapid7, LogRhythm
Challengers	Strong execution but less innovative vision. Reliable for standard needs.	Elastic, LogPoint, NetWitness, Huawei, FireEye, McAfee
Visionaries	Innovative vision but still developing execution capabilities.	Gurukul, Microsoft, Sumo Logic, Fortinet
Niche Players	Limited vision & execution. Serve specific or smaller use cases.	ManageEngine, Odyssey, Micro Focus, Venustech

The report highlights that while Leaders are ideal for large organizations with complex security needs, Challengers can still be effective for organizations with more straightforward requirements.

Log Collection

Log collection is considered one of the most important parts of the SIEM architecture, as without logs the SIEM would be useless. A log is a file that records any event that occurs in a system, whether related to the operating system, other running software, or messages exchanged between users of a communication application. The act of keeping and maintaining these records is called logging.

In its simplest form, a log entry is a single line written to a log file containing four basic elements: the **date and time**, the **source system**, the **process ID**, and a **message** describing what happened.

Example

the /var/log/auth.log file on an Ubuntu server records authentication event, showing entries such as a CRON job opening and closing a session for the root user. Each line provides enough context for a SOC analyst to trace exactly what occurred, when, and on which system.

```
Feb  7 19:15:01 ubuntu CRON[2500]: pam_unix(cron:session): session opened for user root(uid=0) by (uid=0)
Feb  7 19:15:01 ubuntu CRON[2500]: pam_unix(cron:session): session closed for user root
Feb  7 19:17:01 ubuntu CRON[3923]: pam_unix(cron:session): session opened for user root(uid=0) by (uid=0)
Feb  7 19:17:01 ubuntu CRON[3923]: pam_unix(cron:session): session closed for user root
```

Figure 2: Example log entries from /var/log/auth.log

Log entries illustrations

Field	Example Value	Meaning
Date & Time	Feb 7 19:15:01	When the event occurred
Source System	Ubuntu	The machine that generated the log
Process	CRON[2500]	The program that triggered the event
Message	Session opened for user root	What actually happened

Our goal is to transfer logs from various places like hosts, firewalls, proxies, and etc to the SIEM to be able to process all data and detect threats from a central point.

There are 2 ways to collect logs:

- 1- Log Agents
- 2- Agentless

Log Agents

To implement this way, a log agent software is needed.

Agents usually have parsing, log rotation, buffering, log integrity, encryption, conversion features. Meaning, this agent software can take action on the logs it collects before forwarding them to the target.

Example

The agent can divide a log with "username: LetsDefend; account: Administrator" into 2 parts and forward it as:

- message1 = "username: LetsDefend"
- message2 = "account: Administrator"

Pros

- 1- It is a tested, and a working application by the developers.
- 2- Has many additional features like automatic parsing, encryption, log integrity, etc.

Cons

By adding more features, resource consumption increases. That requires the system's resources such as CPU, RAM to be increased, so the cost increases.

Syslog

- A very popular network protocol for log transfers.
- Works with both UDP where the maximum packet size that can be sent with Syslog is 1024 bytes and TCP where the maximum packet size that can be sent with Syslog is 4096 bytes and can be encrypted with TLS.
- Various network and endpoint devices can become syslog-supported with the installation of additional software, including switches, routers, IDS (Intrusion Detection Systems), firewalls, and operating systems such as Linux, Mac, and Windows.
- Log agents can transfer logs using Syslog but logs must be parsed in syslog format

Syslog Format

Timestamp - Source Device - Facility - Severity - Message Number - Message Text

The figure below clearly shows the Syslog format

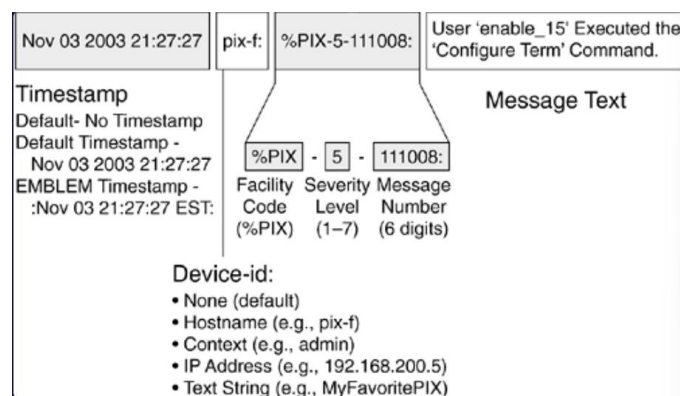


Figure 3: Syslog format

Third-Party Agents

Most SIEM products have their own agent software. But third-party agents may be required as they have more capabilities than syslog due to the extra features they support.

Some agents include:

- Splunk: Universal Forwarder
 - ArcSight: ArcSight Connectors
- ➔ These agents are easy to integrate into SIEM and have parsing features.

Popular open-source agents:

- 1- Beats: <https://www.elastic.co/beats/>
- 2- NXLog: <https://nxlog.co/>

Agentless

- Unlike the agent-based method, agentless log collection requires no software installation on the source device, which reduces both setup time and maintenance costs. Logs are transmitted to the SIEM by connecting directly via SSH or WMI. While this approach is simpler to manage, it carries a security risk since login credentials must be provided, making them potentially vulnerable to theft.
- For this method, the username and password of the log server are required, therefore there is a risk of the password being stolen. Easier to prepare and manage than the agent method. However, it has limited capabilities and credentials are wrapped in the network.

Manual Collection

In some cases, neither agents nor agentless methods can collect logs from certain systems, such as custom cloud-based applications. In these situations, analysts may need to develop their own scripts to extract and forward logs manually to the SIEM.

Find the best log collection method for you Form

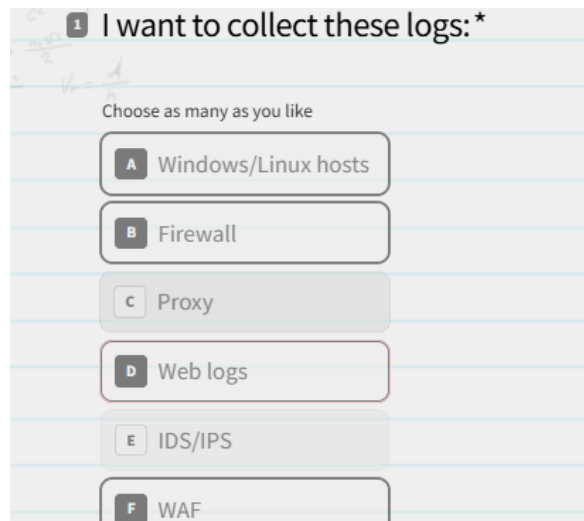
The form helps illustrating the difference between the types of logs collectors.

It started by asking what logs I want to collect (I could choose one or more):

The choices were:

- A- Windows/Linux hosts
- B- Firewall
- C- Proxy
- D- Web Logs
- E- IDS/IPS
- F- WAF
- G- Other

I chose Windows/Linux hosts, Firewall, and Waf as shown in the screenshot below.



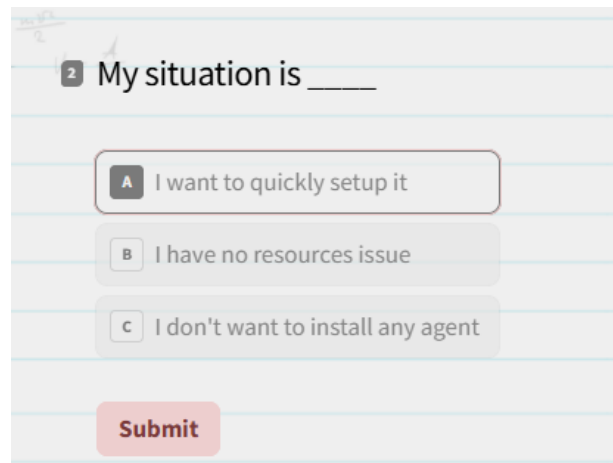
1 I want to collect these logs: *

Choose as many as you like

- ☒ A Windows/Linux hosts
- ☒ B Firewall
- ☐ C Proxy
- ☐ D Web logs
- ☐ E IDS/IPS
- ☒ F WAF

Screenshot 1: Choosing the logs

Then it asked what my situation is like if I want a quick setup or I have no resource bottleneck or I don't want to install an agent



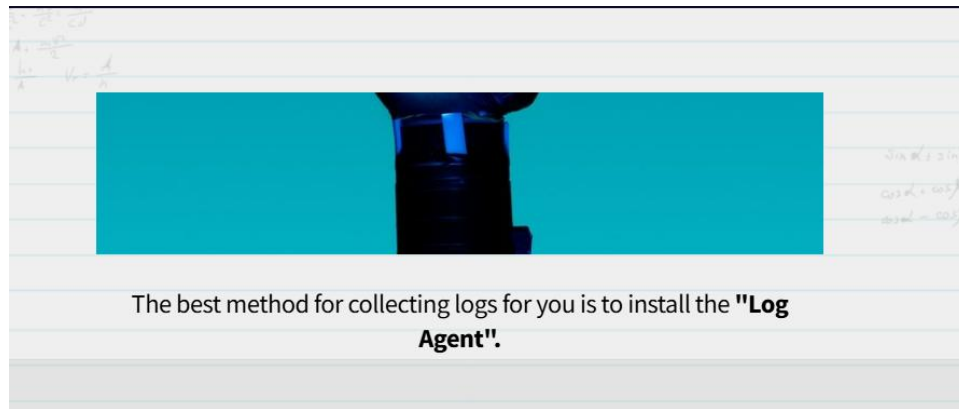
2 My situation is ____

- ☒ A I want to quickly setup it
- ☐ B I have no resources issue
- ☐ C I don't want to install any agent

Submit

Screenshot 2: Deciding my situation

Then the form gave me the agent type that best complies with my choices which was **Log Agent**.



Screenshot 3: The form's response

Log Collection Room Questions

Question 1: What is the best method for those who do not want to manage an agent software?

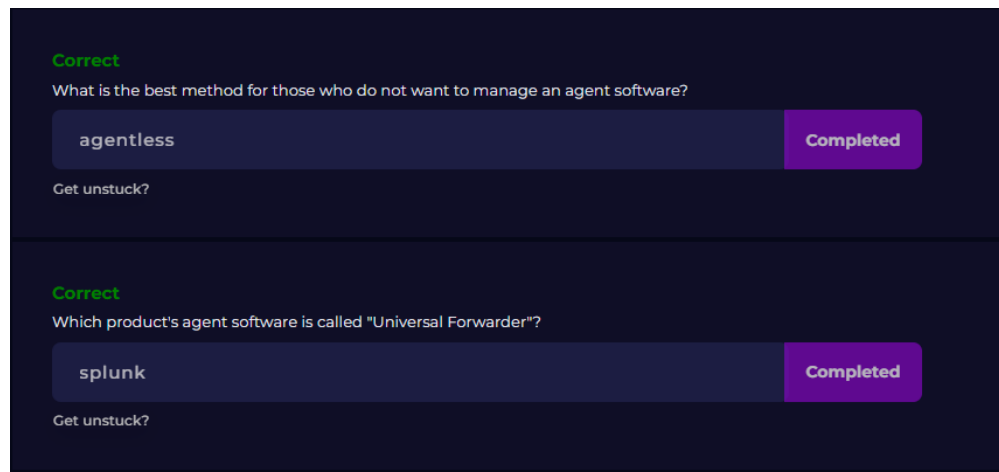
Answer: Agentless.

Justification: The agentless method is the best choice for those who do not want to deal with managing agent software because it requires no installation or maintenance on the source device. Logs are collected by connecting directly to the target machine using protocols like SSH or WMI, which means there is no agent to update, monitor, or troubleshoot. This makes it a simpler and lower maintenance option compared to the log agent method.

Question 2: Which product's agent software is called "Universal Forwarder"?

Answer: Splunk.

Justification: As covered in the Third-Party Agents section of the course, Splunk has its own dedicated agent software called the Universal Forwarder. It is used to collect logs from various sources and forward them to the Splunk server for indexing and analysis. It was specifically mentioned alongside ArcSight Connectors as one of the most commonly used third-party agents in SIEM environments because of how easy it is to integrate and its built-in parsing features.



Screenshot 4: Log Collection section questions and answers

Log Aggregation and Parsing

The log aggregator is where logs are sent to at first. Once logs reach the aggregator, they undergo **Log Processing**. This stage is crucial for efficiency, as it allows for **Filtering**, the process of extracting specific details (such as HTTP status codes) and discarding unnecessary data before it reaches the final destination

This shows that we can filter the logs and send the desired parts to the destination.

The figure below illustrates the parsing process within a SIEM. Raw log entries arriving from various sources are first collected by a log aggregator, which then parses them; breaking the unstructured text into individual labeled fields. This allows the SIEM to efficiently search, filter, and analyze specific values such as IP addresses, timestamps, or HTTP response codes.

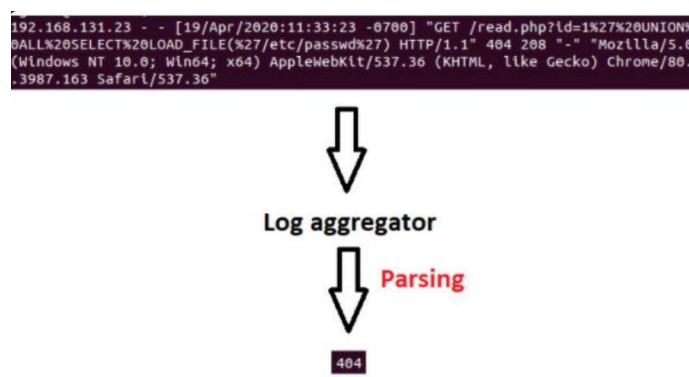


Figure 4: log parsing process inside a SIEM

Parser activity

The room included a practical activity to demonstrate the function of a log parser. The interface provided five distinct log fields and a central **Parser** block. The objective was to drag each field into the parser to see how it categorizes and organizes raw data into structured fields.

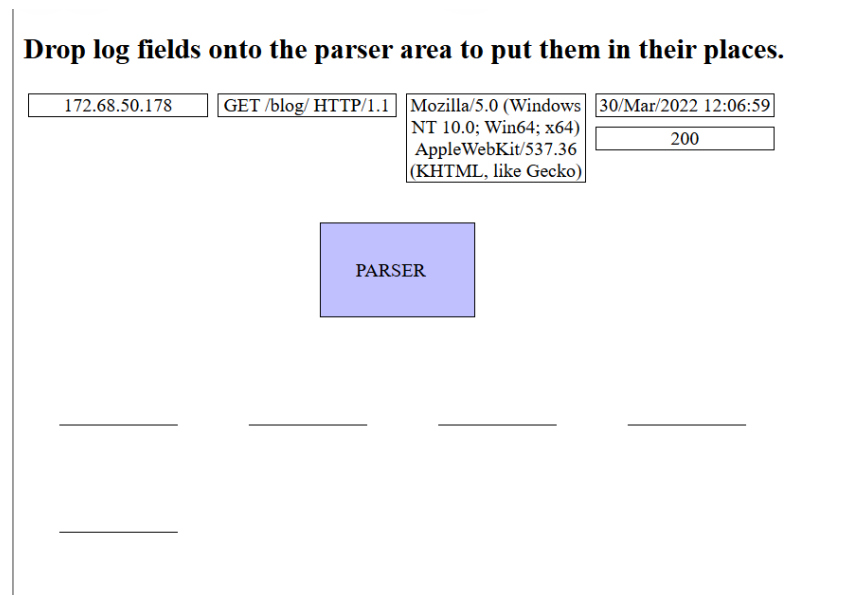


Figure 5: Parser activity

I processed the log entries in the following order:

- **Source IP:** I dragged the 172.68.50.178 block to the parser, which identified and placed it in the **first** slot.
- **Timestamp:** I dragged the 30/Mar/2022 12:06:59 block, representing the date and time, which was positioned in the **second** slot.
- **HTTP Request:** The GET /blog/ HTTP/1.1 block was recognized as the request method and path, then moved to the **third** slot.
- **Status Code:** The 200 block (indicating a successful request) was placed in the **fourth** slot.
- **User Agent:** Finally, the long string containing browser and OS information (Mozilla/5.0...) was dragged to the parser and located in the **fifth** slot.

Drop log fields onto the parser area to put them in their places.

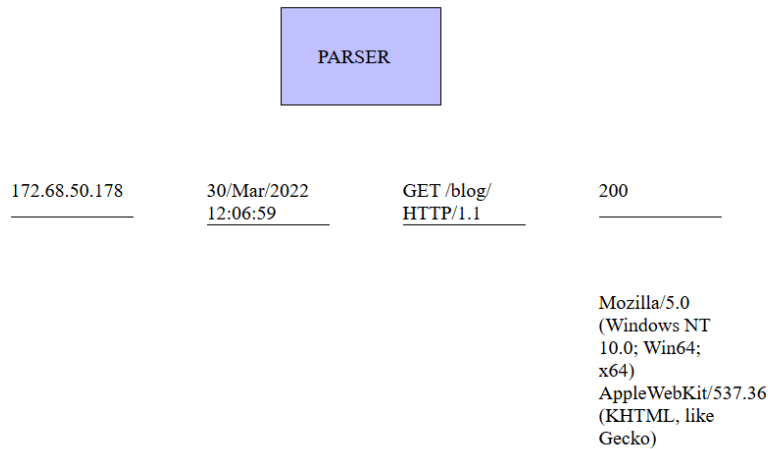


Figure 6: Parser activity solution

Note

The activity demonstrates the process of **Data Normalization**. In a real-world SIEM environment, logs arrive from various sources (firewalls, web servers, databases) in different formats. The **Parser** acts as the translator, extracting key information such as the **Source IP**, **Request Method**, and **Status Code** and mapping them to standard fields. This normalization is critical because it allows security analysts to run uniform queries across different log sources simultaneously.

Aggregator EPS

Events Per Second (EPS)

EPS, which stands for events per second, is a critical metric for scaling SIEM infrastructure. It measures the volume of incoming data using the formula:

$$EPS = \frac{\text{Number of events/logs}}{\text{Time period (seconds)}}$$

So, for example, if the system receives 12000 logs in 10 seconds so the EPS would be $\frac{12000}{10}$ which is equal to 1200

➔ With increasing the EPS value, the aggregator and storage capacity increases.

Log Aggregator

It is a centralized system where logs from various sources are first collected and processed before being forwarded to their final destination.

Scaling the Aggregator

To prevent the logs from overloading the same single aggregator, more than 1 aggregator is added. And sequential or random selection can be provided.

As illustrated in the figure below.

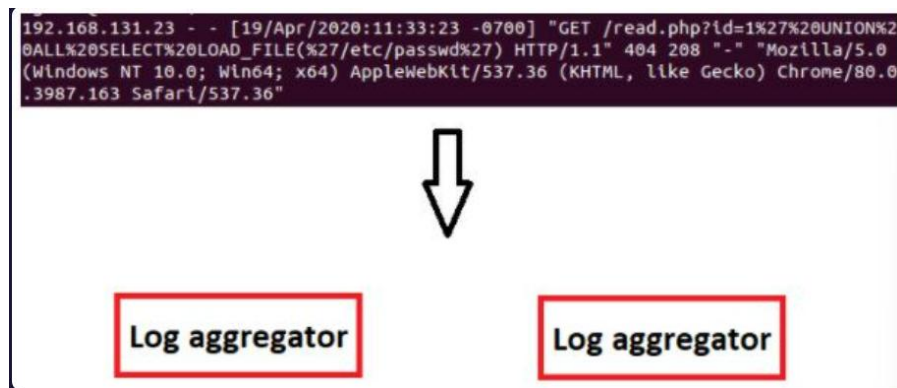


Figure 7: Scaling the aggregator

Log Aggregator Process

When the log arrives at the Aggregator, it goes through one or more process which can be parsing, filtering, and enrichment and then sent to the target.

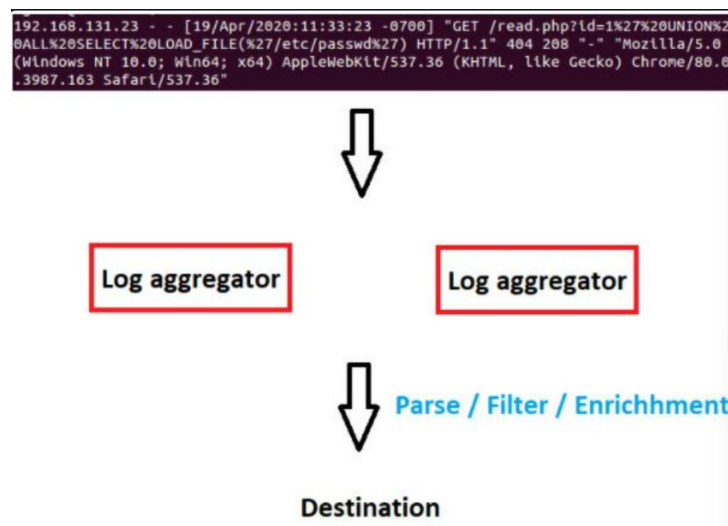


Figure 8: Log aggregator process

Log Modification

Sometimes editing/modifying the log will be needed.

Example the date information of most logs collected comes in the format (day – month – year), but the logs that come from a specific source comes in the format (month – day – year), Therefore, adjusting the format would be needed

Example converting UTC + 2 incoming time information to UTC + 1 would be needed.

Log Enrichment

Enrichment is important as it increases the collected logs' efficiency and saves the analyst's time

Some ways to implement Log Enrichment:

1- Geolocation:

Facilitates the process on the person viewing the log and saves their time. Also allows analyzing location-based behavior.

Example Adding the geolocation of the specified IP address to the log

2- DNS:

Since DNS queries helps mapping IP addresses to its the domain. Therefore, an IP address can be found using reverse DNS

3- Add/Remove:

Means the aggregator can automatically add extra details to a log to make it more useful or take out unnecessary parts to keep the file size small and clean.

Log Aggregation and Parsing section quiz

Question 1: Which one is not the function of a log aggregator?

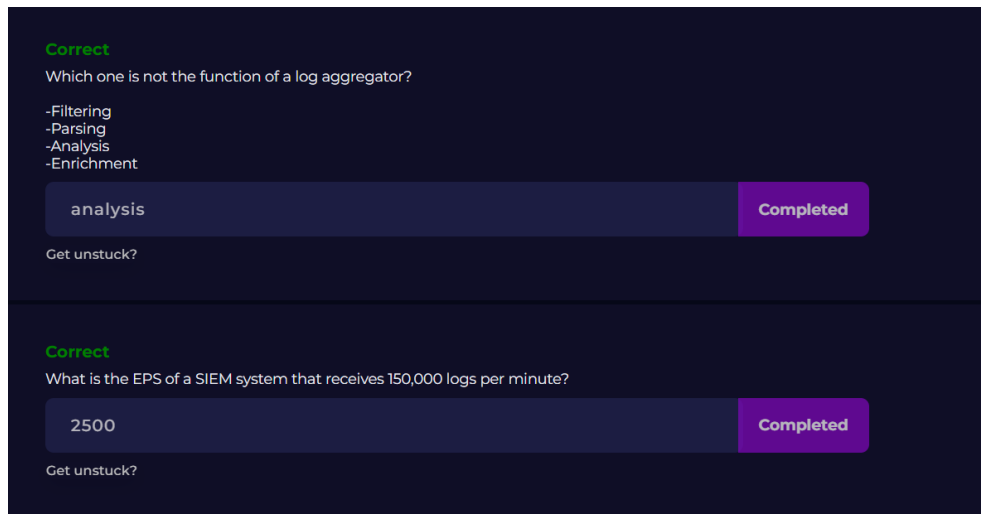
- Filtering
- Parsing
- Analysis
- Enrichment

Answer: Analysis.

Justification: While an aggregator handles **Parsing**, **Filtering**, and **Enrichment**, "Analysis" is the human-led or algorithmic process that occurs *after* the data has been normalized and stored in the SIEM.

Question 2: What is the EPS of a SIEM system that receives 150,000 logs per minute?

Answer Using the $EPS = \frac{\text{Number of events/logs}}{\text{Time period (seconds)}}$ rule. The answer will be $\frac{150000}{1 \text{ minute} * 60 \text{ seconds}} = 2500$



Screenshot 5: Log Aggregation and Parsing section questions and answers

Log Storage

Following the log aggregation and normalization process, the processed data is forwarded to **Storage** for retention and future security analysis.

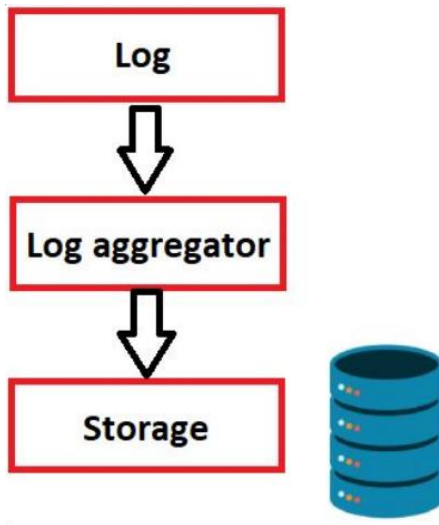


Figure 9: The log data flow (The storage phase)

A common pitfall in SIEM design is over-prioritizing **Storage Capacity** while neglecting **Search Performance**. While maintaining a large historical archive is necessary, the speed of data retrieval is equally critical for effective incident response.

Scenario

If a SOC analyst searches through logs from WAF, Firewalls, and Proxies and the query takes 15 minutes to complete, the investigation becomes unproductive. High-latency data access significantly hinders a SOC's ability to respond to active threats in real-time.

WORM Technology and Indexing

Standard relational databases (like MySQL) are optimized for frequent adding, editing, and deleting of data. However, SIEM storage focuses on **Indexing** for rapid search and retrieval. Because security logs should never be modified once recorded, SIEMs utilize **WORM (Write Once, Read Many)** based technologies. This approach ensures:

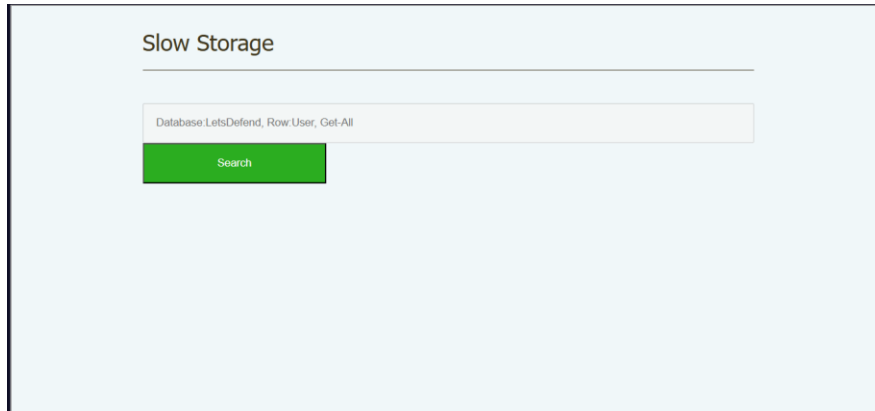
- **Data Integrity:** Logs remain in their original state for forensic purposes.
- **High-Speed Access:** The system is optimized to search and "read" millions of events as quickly as possible.

➔ To know further information about WORM (write once read many):

https://en.wikipedia.org/wiki/Write_once_read_many

Slow storage VS Fast storage practical activity

I was given a slow storage database where there was a search button as shown below



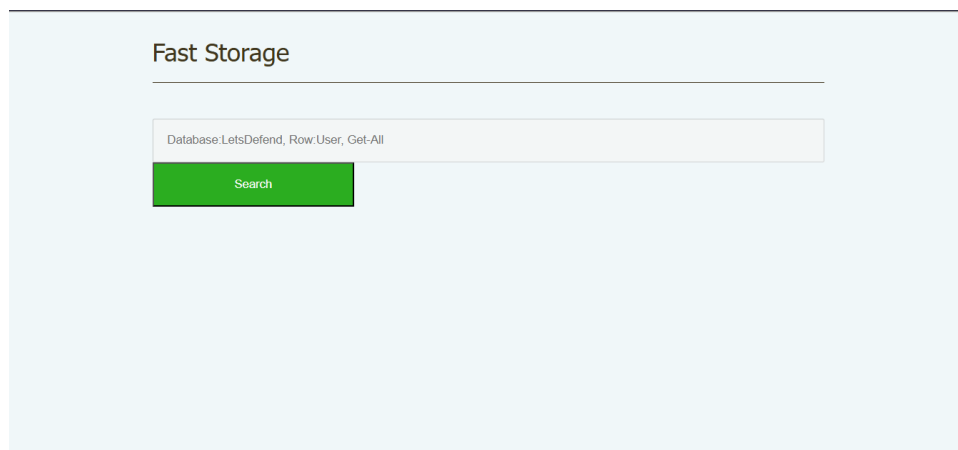
Screenshot 6: Slow storage before searching

I pressed the search button and after a few seconds, a table of data was shown as shown in the screenshot below.

Name	Surname	Age	Id
Joe	Allen	33	1
Alan	Kowalsky	63	2
Victor	Scendell	15	3
Jeniffer	Lescott	44	4
Paul	Finley	70	5
John	Batshuai	23	6

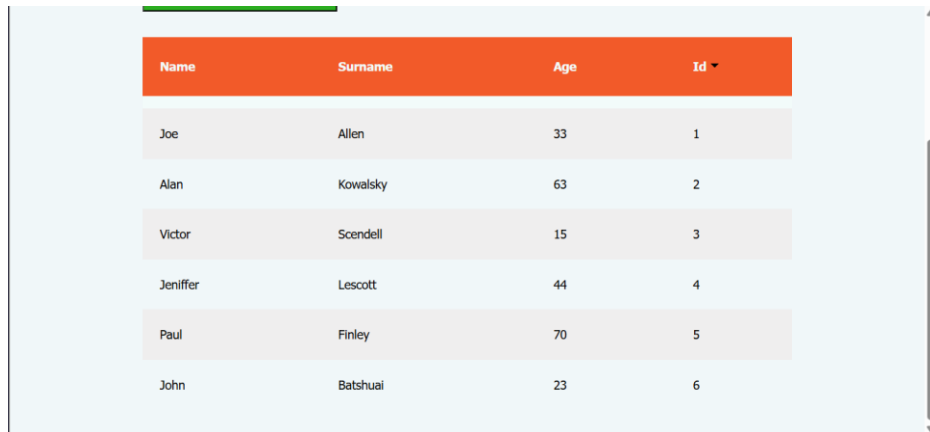
Screenshot 7: Slow storage after searching

Then I was given a fast storage database where there was also a search button as shown below.



Screenshot 8: Fast storage before searching

I pressed the search button a table of data was shown instantly.



Name	Surname	Age	Id
Joe	Allen	33	1
Alan	Kowalsky	63	2
Victor	Scendell	15	3
Jeniffer	Lescott	44	4
Paul	Finley	70	5
John	Batshuai	23	6

Screenshot 9: Fast storage after searching

Log Storage Room Questions

Question 1: Is updating data (changing values, deleting values, etc.) very important for SIEM data storage? Answer Format: Y/N

Answer: N

Justification: Unlike traditional databases (such as MySQL) that prioritize the ability to edit or delete records, SIEM storage is built on the **WORM (Write Once, Read Many)** principle. In a security context, log data must remain immutable to ensure **data integrity** for forensic investigations. The primary focus of SIEM storage is not modification, but **indexing**; enabling the system to search and retrieve millions of events as quickly as possible without ever changing the original log entries.

Question 2: Which one is the most important for SIEM storage?

- Speed
- Features
- Price

Answer: Speed

Justification: While storage capacity and cost are relevant, **speed** is the most critical factor because it directly impacts the efficiency of security operations. In a SOC environment, analysts must be able to query massive volumes of data (such as WAF, Firewall, and Proxy logs) and receive results instantly. So for example, if data retrieval is slow, taking 15 minutes for a single search, the investigation becomes unproductive, and the ability to respond to active threats in real-time is compromised. Therefore, SIEM storage must be optimized for high-speed indexing and rapid data access.

Correct
Is updating data (changing values, deleting values, etc.) very important for SIEM data storage?
Answer Format: Y/N
N Completed
Get unstuck?

Correct
Which one is the most important for SIEM storage?
-Speed
-Features
-Price
speed Completed
Get unstuck?

Screenshot 10: Log Storage section questions and answers

Alerting

Following the log retention phase in **Storage**, the SIEM moves into the **Alerting** stage. In this final step of the lifecycle, the system analyzes the normalized and indexed data to identify patterns that match known threat signatures or anomalous behavior, automatically generating notifications for the SOC team to investigate.

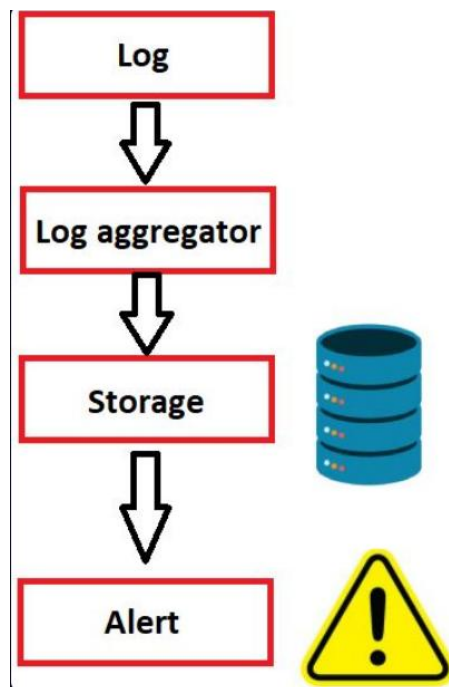


Figure 9: The log data flow (The alerting phase)

Alerting Strategy

The timing of an **alert** depends on how fast your storage can be searched. If something bad happens today, we want to know about it right now, not two days from now. This is why having fast, well-organized storage is so important.

Methods of Creating Alerts

There are two main ways the SIEM makes an alarm:

- 1- **Searching Saved Data:** The system looks through the logs it already stored to find patterns.
- 2- **Real-time Alerts:** The system creates a warning the exact moment it receives a log.

Keeping Alerts Useful

A good alert should only go off when something needs to be investigated. If a SIEM sends too many alarms, the analysts will get overwhelmed.

Examples of helpful alerts

- **Admin Changes:** A warning when someone is added to the "Global Administrator" group.
- **Brute Force:** A warning when there are **15 failed logins in 3 minutes** from the same user.

How to Search Better

To make high-quality alerts, you need to use **these three search tricks:**

a. Blacklisting

Blacklisting is a straightforward method used to detect known malicious activity. By maintaining a list of prohibited items, such as specific process names (e.g., mimikatz.exe) or banned IP addresses. The SIEM can trigger an **alert** the moment a match is found in the logs.

Pros

- **Ease of Use:** It is easy to implement and manage within a SIEM environment.
- **Known Threat Detection:** Highly effective at stopping specific, previously identified malicious actors or tools.

Cons

- **Easily Bypassed:** This method is very easy for attackers to circumvent.
- **Exact Match Reliance:** If a threat actor simply renames a malicious file to mimikatz2.exe, the blacklist rule will not recognize it, and no alert will occur.

Example

If a threat actor renames a malicious file to mimikatz2.exe, the blacklist rule will not recognize it, and no alert will occur. This highlights why blacklisting should be just one part of a broader detection strategy.

b. Whitelisting

Unlike blacklisting, **whitelisting** is used to define desired or safe behavior. For example, an analyst can maintain a list of internal IP addresses that are known to communicate normally. If a device initiates communication with an address that is not on this list, the SIEM can trigger an **alert** for the unauthorized connection.

Pros

- **High Effectiveness:** Highly effective at catching unknown threats.
- **Security Precision:** Ensures only pre-approved activities are permitted, significantly reducing the attack surface.

Cons

- **Management Complexity:** It is difficult to manage in large environments.
- **Maintenance Requirements:** Because networks change frequently, the whitelist must be constantly updated to avoid generating false alarms for legitimate new systems.

c. Long Tail Analysis

This detection method is based on the principle that high-frequency behaviors are typically normal, while rare events are suspicious. For example, if a device constantly generates **Event ID 4624 (An account was successfully logged on)**, this pattern is treated as a baseline of normal activity. Under **Long Tail Analysis**, an analyst focuses their attention on the "least occurring" logs, the events that happen only once or twice across the entire network.

Pros

- **Effective at identifying stealthy threats:** Because advanced attackers often try to stay quiet, their actions frequently appear as rare outliers rather than common, high-volume events.
- **Discovers unknown anomalies:** It helps find suspicious activities that wouldn't be caught by a standard blacklist.

Cons

- **Time-consuming:** Analyzing the "tail" of the data can be labor-intensive for an analyst to filter through manually.
- **Potential for noise:** Not every rare event is malicious; some may just be infrequent but legitimate administrative tasks.

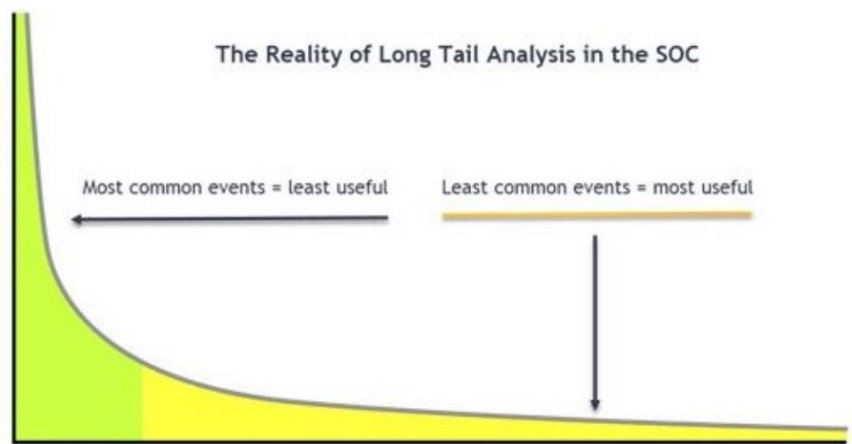


Figure 10: The Concept of Long Tail Analysis in the SOC

Source: <https://respond-software.com/>

This figure above illustrates the statistical distribution of log events and their value to a security analyst:

- **The Common Peak (Left):** Represents high-volume, repetitive logs that are typically "least useful" because they reflect routine, normal system behavior.
- **The Long Tail (Right):** Represents the "least common events" which are "most useful" to an analyst.
- **Analytic Value:** In a SOC environment, stealthy or advanced threats often appear as single, rare outliers (the "tail") rather than frequent events.

Summary table to differentiate between the techniques

Technique	Core Logic	Pro	Con
Blacklisting	Block known "Bad"	Easy to implement	Easy to bypass (renaming files)
Whitelisting	Allow known "Good"	Highly effective/secure	High maintenance; needs constant updates
Long Tail	Investigate the "Rare"	Catches stealthy attackers	Can be time-consuming to filter manually

Alerting section quiz

Question 1: There are two IP addresses that are definitely malicious. Which method should be used to create an alert when these IP addresses are accessed?

- Whitelisting
- Blacklist
- Long tail

Answer: Blacklist

Justification: Blacklisting is the most straightforward method for detecting known malicious activity. Since the two IP addresses are already identified as "definitely malicious," they can be added to a prohibited list. The SIEM will then trigger an immediate alert whenever a match is found in the incoming logs.

Question 2: "The whitelist method is effective and easy to manage." Is this statement true or false?

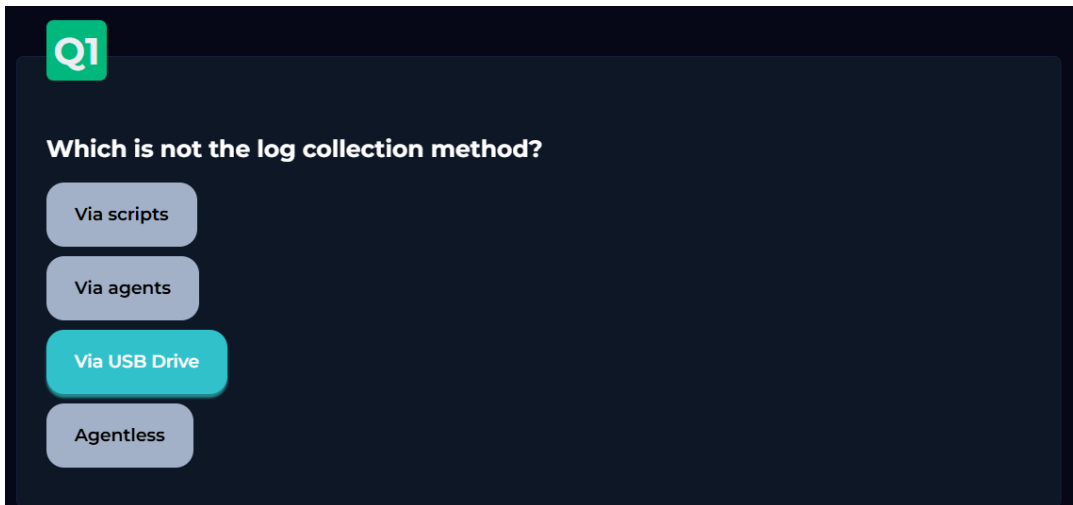
Answer: False

Justification: While the whitelist method is highly effective because it defines exactly what communication is "normal," it is not easy to manage. This method requires constant manual updates to include new legitimate devices or services. In large, frequently changing environments, the high administrative effort needed to prevent false alarms makes it a difficult strategy to maintain.

The screenshot displays a quiz interface with a dark background. The first question is: "There are two IP addresses that are definitely malicious. Which method should be used to create an alert when these IP addresses are accessed?". The options are "-Whitelisting", "-Blacklist", and "-Long tail". The answer "blacklist" is entered in a text field, and a purple "Completed" button is visible. Below the answer field is a link that says "Get unstuck?". The second question is: "'The whitelist method is effective and easy to manage.' Is this statement true or false?". The answer "false" is entered in a text field, and a purple "Completed" button is visible. Below the answer field is a link that says "Get unstuck?".

Screenshot 11: Alerting section questions and answers

SIEM Introduction room quiz



Screenshot 12: SIEM Introduction room quiz question 1

Question 1: Which is not the log collection method?

Answer: Via USB Drive

Justification: Standard SIEM architecture relies on automated and scalable log ingestion methods to ensure real-time monitoring. These include:

- **Via Agents:** Installing dedicated software on endpoints for advanced processing.
- **Agentless:** Using network protocols like SSH or WMI to pull data without local installation.
- **Via Scripts:** Developing custom scripts (manual collection) for unique or cloud-based applications.

Collecting logs **Via USB Drive** is not a recognized SIEM collection method because it is manual, not scalable, and cannot provide the real-time data flow necessary for modern threat detection.

Q2

Which one is not the cons of agentless log collection?

Difficult to monitoring agent health

Not possible to implement filter

No required log collection software

Need SSH, WMI or something like that

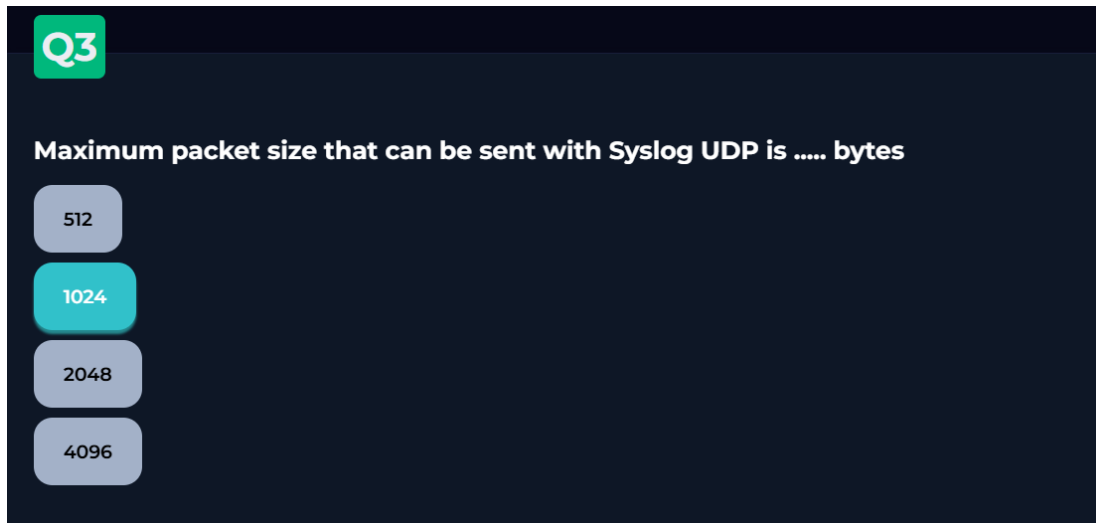
Screenshot 13: SIEM Introduction room quiz question 2

Question 2: Which one is not the cons of agentless log collection?

Answer: No required log collection software

Justification: The primary advantage (pro) of agentless log collection is that it does not require installing additional software on each target device, which reduces the administrative burden and preserves system resources. The other options listed are actual disadvantages (cons) of this method:

- **Difficult to monitor agent health:** Since there is no agent software, you cannot easily verify the "health" or status of the collection process on the endpoint.
- **Not possible to implement filter:** Processing and filtering typically happen at the destination rather than the source, potentially increasing network traffic.
- **Need SSH, WMI, or something like that:** This method relies on existing network protocols and permissions, which can create security risks if these services are not properly hardened.

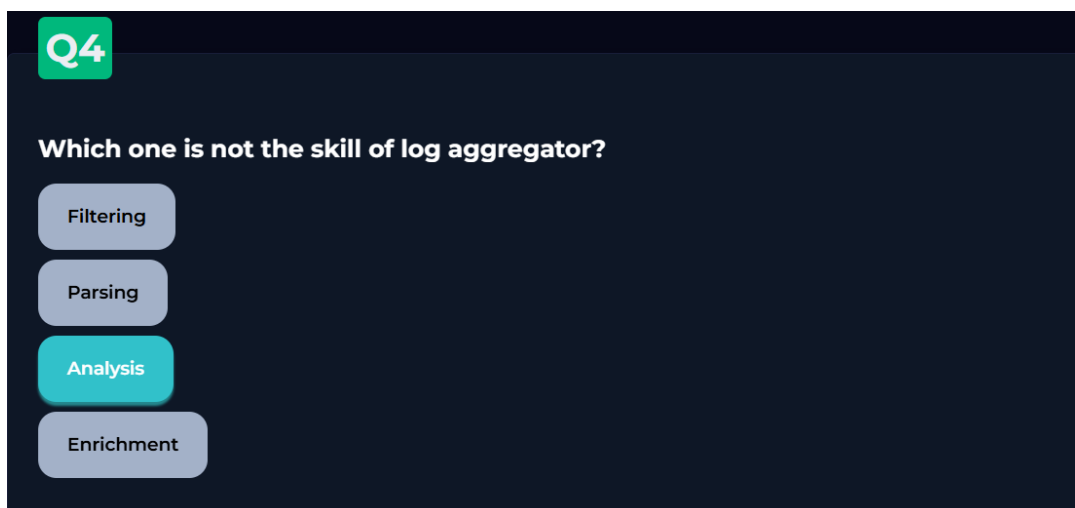


Screenshot 14: SIEM Introduction room quiz question 3

Question 3: Maximum packet size that can be sent with Syslog UDP is bytes

Answer: 1024

Justification: According to the traditional RFC 3164 standard, which governs the BSD Syslog protocol, the total size of a Syslog packet sent over UDP is limited to 1024 bytes. This limitation ensures that the message can be transmitted across various network types without being fragmented. While newer standards (like RFC 5424) allow for larger sizes, the 1024-byte limit remains the foundational standard for basic Syslog UDP implementations in SIEM log collection.



Screenshot 15: SIEM Introduction room quiz question 4

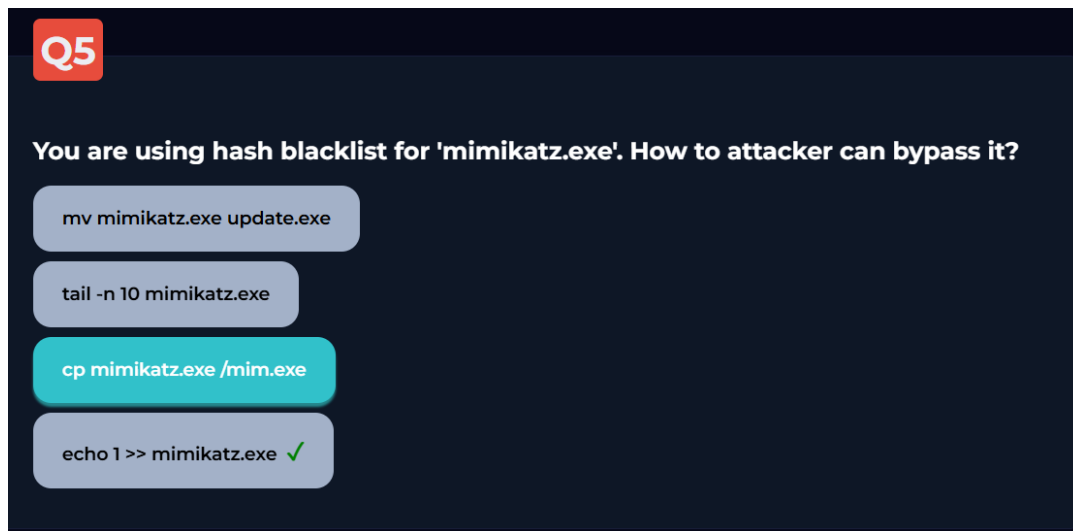
Question 4: Which one is not the skill of log aggregator?

Answer: Analysis

Justification: A log aggregator is designed to prepare data for the SIEM, not to perform the final security investigation. Its primary functions, or "skills," include:

- **Filtering:** Removing unnecessary data to save storage space.
- **Parsing:** Breaking down raw text into structured, readable fields.
- **Enrichment:** Adding extra context (like Geo-IP or user details) to the logs.

Analysis is not a function of the aggregator; rather, it is the step that happens *after* the logs have been aggregated and stored. Analysis is performed by the SIEM's correlation engine or by SOC analysts using the processed data to detect threats.



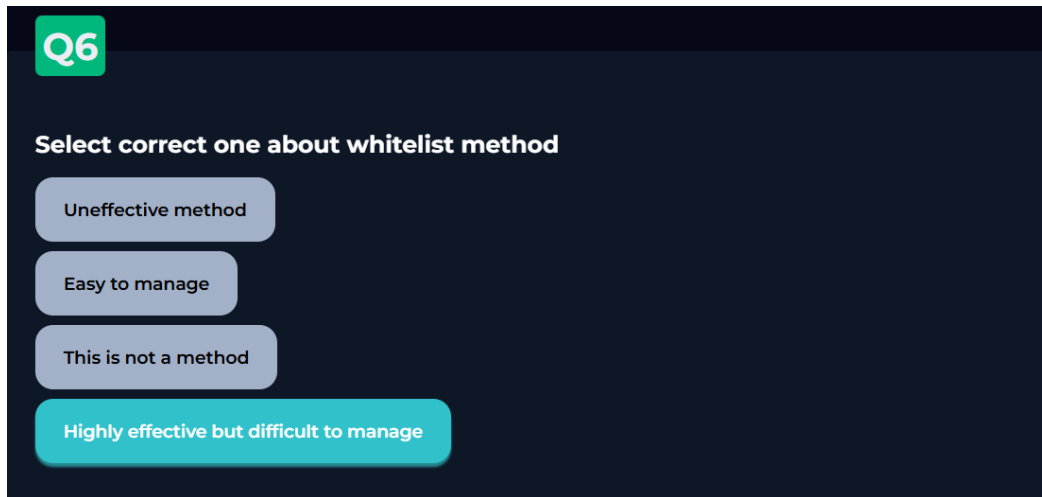
Screenshot 16: SIEM Introduction room quiz question 5

Question 5: You are using a hash blacklist for 'mimikatz.exe'. How can an attacker bypass it?

Answer: echo 1 >> mimikatz.exe

Justification & Lessons Learned: In my initial analysis of this question, I considered renaming the file (Option 3: cp mimikatz.exe /mim.exe) as the bypass. However, through the "SIEM Introduction" lab, I learned that while renaming works for a **name-based blacklist**, it is ineffective against a **hash-based blacklist**.

A hash is a unique digital signature of a file's contents. Simply renaming a file does not change its hash. To successfully bypass a hash-based blacklist, the attacker must alter the file's internal data. By using the command echo 1 >> mimikatz.exe, the attacker appends a character to the file, which fundamentally changes its hash. This results in a "new" file that the SIEM's blacklist will not recognize, allowing the malicious tool to bypass detection.

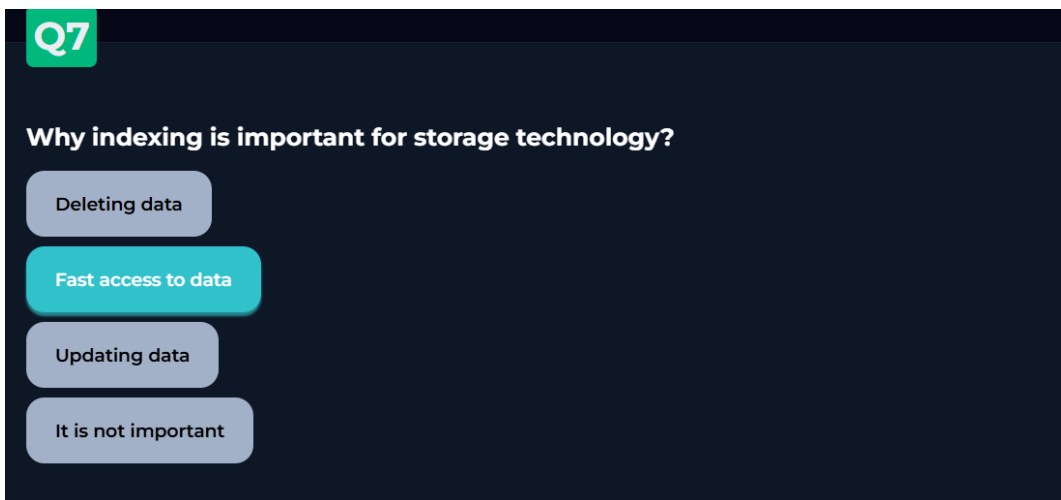


Screenshot 17: SIEM Introduction room quiz question 6

Question 6: Select the correct statement about the whitelist method.

Answer: Highly effective but difficult to manage

Justification: Whitelisting is considered one of the most secure detection methods because it follows a "deny-by-default" logic, allowing only pre-approved activities or connections. This makes it **highly effective** at stopping unknown threats and "zero-day" attacks. However, it is **difficult to manage** because it requires a deep understanding of the network baseline. In dynamic environments where new software and services are frequently added, the whitelist must be manually updated constantly to prevent a flood of false positive alerts.

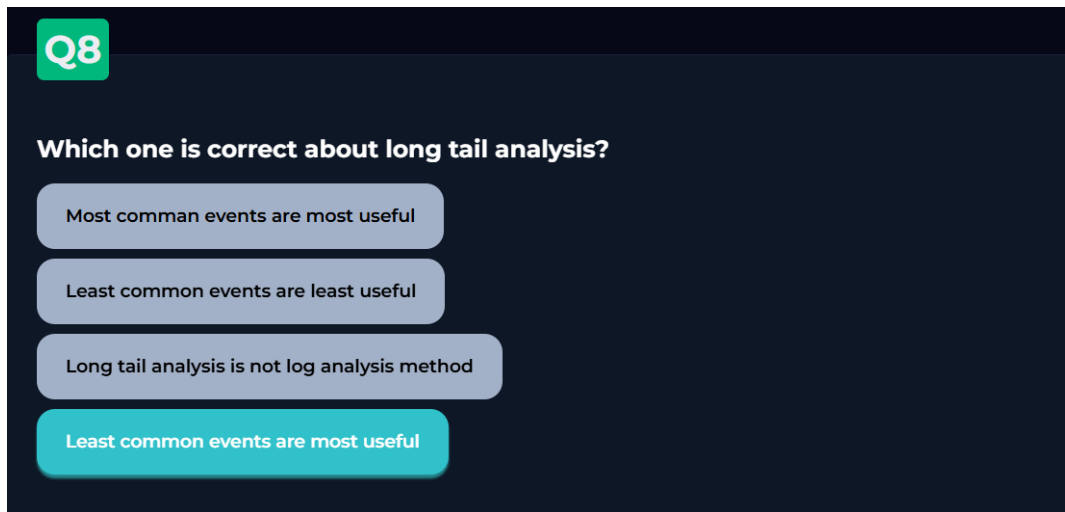


Screenshot 18: SIEM Introduction room quiz question 7

Question 7: Why indexing is important for storage technology?

Answer: Fast access to data

Justification: In the context of a SIEM, storage technology must be optimized for retrieval rather than modification. **Indexing** creates a searchable map of the stored logs, allowing the system to pinpoint specific events across terabytes of data almost instantaneously. Without proper indexing, an analyst might have to wait minutes or even hours for a single query to finish, which is unacceptable during a live security incident where every second counts. Since security logs are rarely updated or deleted, the primary goal of the storage layer is ensuring that access remains as fast as possible.

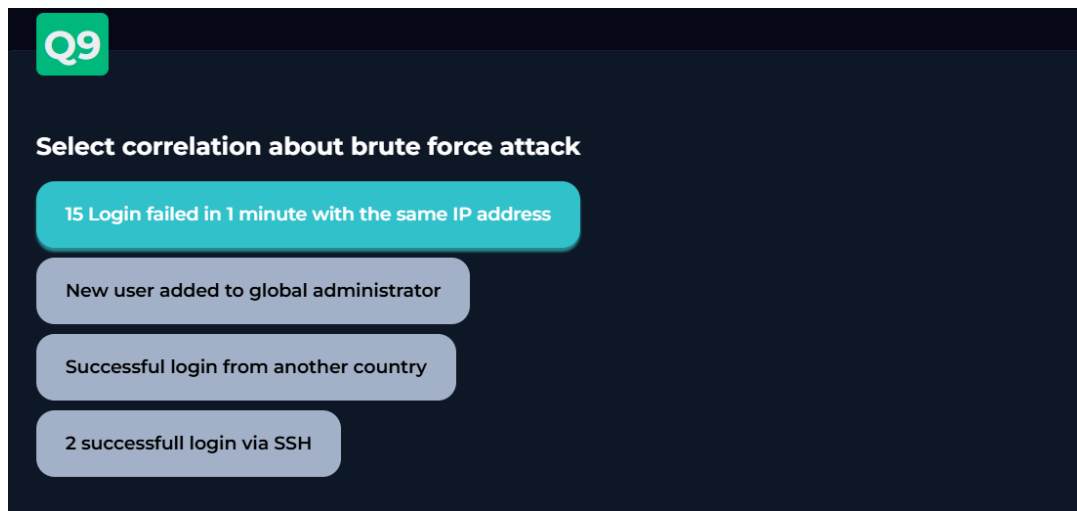


Screenshot 19: SIEM Introduction room quiz question 8

Question 8: Which one is correct about long tail analysis?

Answer: Least common events are most useful

Justification: Long tail analysis operates on the statistical principle that normal network behavior is repetitive and frequent, creating the "peak" of a data distribution. Conversely, security threats, unauthorized access, or stealthy attacker movements are rare and appear as outliers. Therefore, the **least common events**—those that occur only a few times—are considered the **most useful** to a SOC analyst because they are the most likely to contain evidence of a targeted attack that would otherwise be hidden in the noise of high-volume, "common" logs.



Screenshot 20: SIEM Introduction room quiz question 9

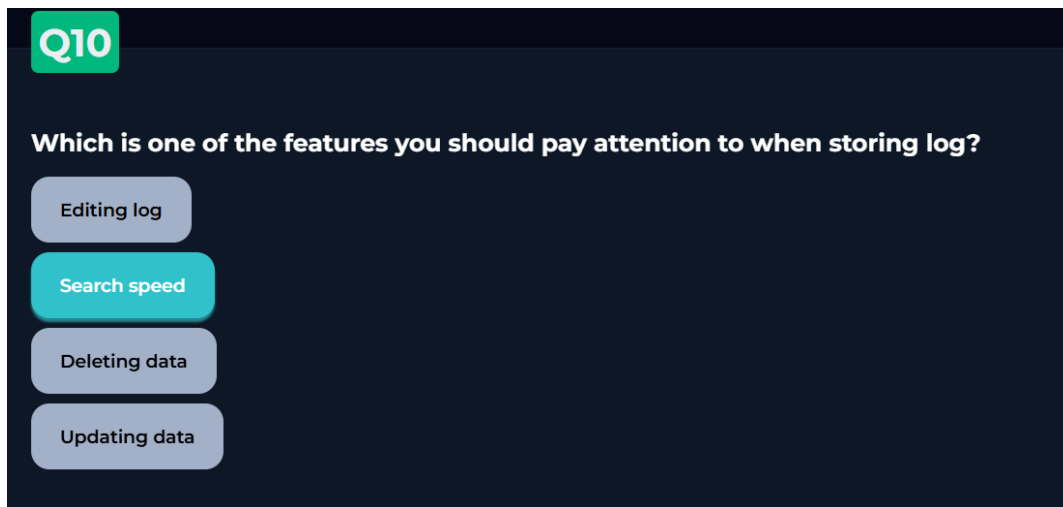
Question 9: Select the correlation about brute force attack.

Answer: 15 Login failed in 1 minute with the same IP address

Justification: A brute force attack involves an automated attempt to guess passwords by trying many combinations in a very short period. To detect this, a SIEM uses **correlation rules** that look for a specific threshold of events. In this case, multiple failed login attempts coming from a single source (the same IP address) within a narrow time window (1 minute).

The other options represent different types of security events:

- **New user added to global administrator:** This is a **Privilege Escalation** alert.
- **Successful login from another country:** This is an **Anomalous Location** or "Impossible Travel" alert.
- **2 successful login via SSH:** This is likely normal administrative activity unless it deviates from a known baseline.



Screenshot 21: SIEM Introduction room quiz question 10

Question 10: Which is one of the features you should pay attention to when storing log?

Answer: Search speed

Justification: When designing a storage environment for a SIEM, **search speed** is the most critical technical factor. In a security operations center (SOC), analysts need to query massive amounts of historical data to track an attacker's movement or investigate a breach. If the storage is slow, the response time is delayed, which can allow a threat to spread further.

The other options are features that should be avoided or restricted in a professional logging environment:

Editing, Updating, and Deleting:

Logs must maintain **Data Integrity**. Modern SIEMs use **WORM** (Write Once, Read Many) technology to ensure that once a log is created, it cannot be tampered with or deleted, protecting the evidence for forensic investigations.

Conclusion

Going through this course made it clear that a SIEM is not just a single tool but rather a full pipeline where every stage depends on the one before it. Without proper log collection there is nothing to analyze, and without fast storage there is no point in having alerts. The course succeeded in breaking down a complex system into understandable stages, giving a realistic picture of what a SOC analyst deals with on a daily basis. Overall, this course provided a solid and well-structured introduction to how modern SOC environments detect and respond to threats using SIEM technology, and understanding these fundamentals is essential for any analyst working in a SOC environment.